Sampling variability

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which units were sampled and what makes each sample equally likely.

Teach exact probabilities without inventing observed studies, independent draws or classifier confidence intervals.

Keep one habit: Model six distinct-ID pairs → inspect probability bars → contrast two classifier rates → assess new-value enumeration → preserve selection and evidence limits.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **sampling variability**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of confounders and controls

Reference example, not the only acceptable wording: Teacher explanation: A confounder changes alongside the intended factor and can offer another explanation. Controls aim to isolate the proposed **mechanism**.

For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

Without opening the old lesson, explain one key idea from Baselines and fair comparisons. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of baselines and fair comparisons

Reference example, not the only acceptable wording: Teacher explanation: Fair baselines use credible procedures on matched **evidence** and resource conditions. A weak arbitrary reference can exaggerate improvement.

For a room-reporting study use manual authoritative lookup or a deterministic same-room rule, not random room names. Match cases, available sources and scoring.

Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of replication and reproduction

Reference example, not the only acceptable wording: Teacher explanation: Reproducing a computation uses recorded artifacts; replication collects new **evidence** under a comparable question.

Naming conventions vary, so define the terms in the project. Here replay means same artifacts/procedure; new-data replication means newly collected eligible cases under the same plan.

Preserve environment, data versions and raw results.

Pictured end goal

Explain **sampling variability** using the records and boundaries of this fictional investigation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Confounders and controls

Baselines and fair comparisons

Replication and reproduction

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. **Sampling variability** means different eligible samples can give different estimates even under the same population. Separate this from procedure changes.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **sampling variability**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach exact sampling possibilities and observed-rate limits separately

> Do not transfer the finite toy model's probabilities to unrelated classifier samples.

Definitions and complete finite-population contract

Sampling variability means different eligible samples can give different estimates even under the same population. Separate this from procedure changes.

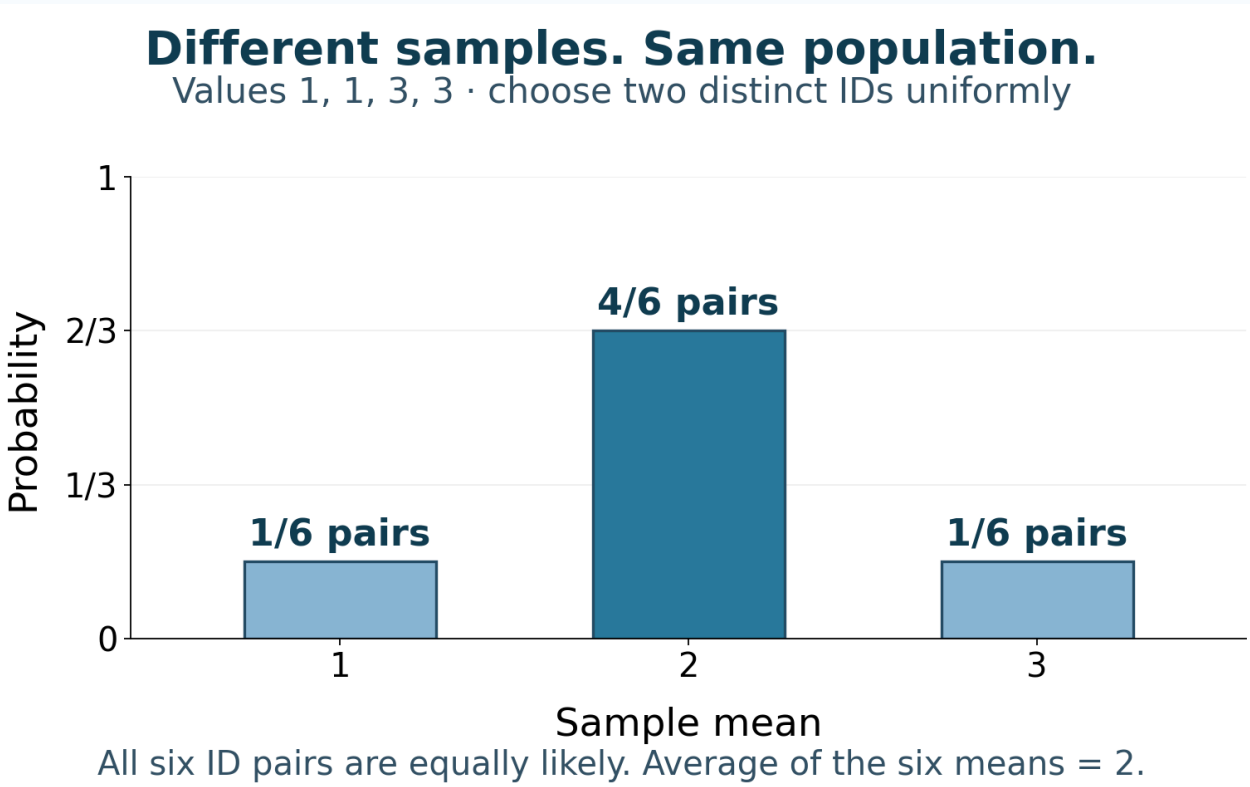
Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 **without replacement**. Their means are 1, 2, 2, 2, 2, 3.

Sampling-model item	Complete fictional finite-population rule
Known population	U1=1, U2=1, U3=3, U4=3; four distinct IDs
Sample	Two different IDs, without replacement within a sample
Probability model	All six unordered ID pairs equally likely
Statistic	Arithmetic mean of the two selected values
All possible sample means	1, 2, 2, 2, 2, 3 in the declared pair order
Probability of mean 2	Four of six eligible pairs = 2/3

Sampling-model item	Complete fictional finite-population rule
Average of all possible means	2, equal to this known population mean
Dependence and counting	Do not merge IDs with equal values; without-replacement draws are not independent
Separate picture	Different four-case classifier samples score 3/4 and 2/4; no sampling probability model supplied
Scope	Exact enumeration under the stated toy model, not a random experiment, confidence interval or real model evaluation

Main ID-pair ledger, native distribution and checked reasoning

Unordered ID pair	Selected values	Sample mean	Model probability
U1, U2	1, 1	1	1/6
U1, U3	1, 3	2	1/6
U1, U4	1, 3	2	1/6
U2, U3	1, 3	2	1/6
U2, U4	1, 3	2	1/6
U3, U4	3, 3	3	1/6
Summary	Six equally likely ID pairs	Average = (1+2+2+2+2+3)/6 = 2	P(mean 2) = 4/6 = 2/3



Main U1/U2 have value one and U3/U4 value three. Six uniform unordered two-ID samples give means one, two, two, two, two, three; average two and probability of mean two is four sixths. Equal values remain distinct IDs, enumeration is not six observed experiments, and without-replacement draws are not independent. Picture rates 75%/50% differ by 25 points, with no supplied sampling model or interval.

Identical fixed deterministic records should agree on replay. Selected values zero, zero, four, four yield means zero, two, two, two, two, four; average two and mean two occurs four of six. Preserve exact IDs and the uniform sample rule, without claiming a real study or broader uncertainty analysis.

Sampling variability — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2.	Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 without replacement.	Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ. Qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2.

Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Selection, identity and deterministic-replay probes

Sampling-variability probe	Expected interpretation
Equal values with distinct IDs	Remain separate eligible units
Choose two uniformly without replacement	Six equally likely unordered pairs for four IDs
Six sample possibilities listed	Not six independently observed experiments
A sample mean is 1 or 3	Compatible with fixed population mean 2 under this model
Four pairs have mean 2	Exact model probability 4/6, not a universal sampling rule
Selection rule changes	Recompute probabilities; do not reuse uniform-pair weights blindly
New classifier sample differs	May reflect changed included cases; not automatic model drift
Same deterministic cases score differently	Inspect data, code, settings and scoring changes
Only two observed rates supplied	No confidence interval or probability of the difference established

Separate classifier samples and model-information contrast

Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection.

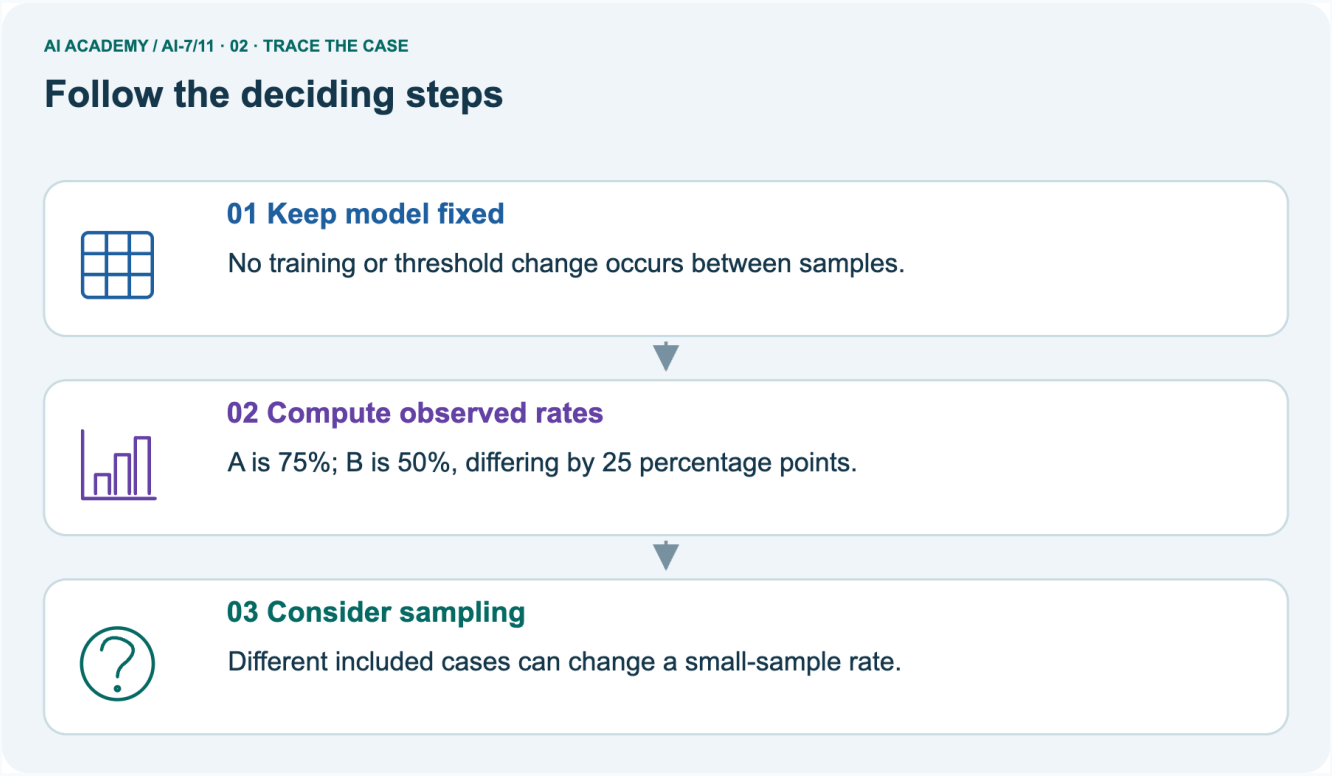
Sample A has 3 correct of 4; sample B has 2 correct of 4. The model and rubric are unchanged.

The picture is a separate fixed-classifier example, not another draw from the four-card population. Sample A has three correct of four, or 75%; B has two of four, or 50%, a difference of 25 percentage points. Different included cases can change the observed rate without changing the classifier or rubric.

But no sampling design or probability model is supplied for these two samples, so do not borrow the main example's 2/3 probability or invent a confidence interval. Repeating the same cases with deterministic fixed predictions and unchanged scoring should give the same count; investigate an unexplained mismatch rather than calling it new-sample variability.

Evidence distinction	Main finite model	Separate classifier picture
Population/sample information	All four IDs/values and uniform pair selection known	Only two four-case counts supplied
Observed/possible quantities	Six possible means enumerated exactly	A 3/4 = 75%; B 2/4 = 50%
Computed probability	P(mean 2)=2/3 under stated model	No probability model supplied
What changes	Which two IDs enter a sample	Which four cases enter the scored sample
Identical deterministic replay	Same chosen records give same mean	Same cases/predictions/scoring give same rate
Not established	Broader real-population inference	Cause, population precision or confidence interval

Picture trace and identical-record replay key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep model fixed	No training or threshold change occurs between samples.
2. Compute observed rates	A is 75%; B is 50%, differing by 25 percentage points.
3. Consider sampling	Different included cases can change a small-sample rate.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

These two samples do not supply a probability model or confidence interval; none is invented.

Change: Evaluate the exact same fixed four cases twice with deterministic **prediction**s.

Prediction: The counts should agree unless some condition, data or scoring changed.

Reason: The sampling explanation no longer applies when the records are identical.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Sampling variability	Changes in estimates across eligible samples
Sampling distribution	The distribution of an estimate under a sampling procedure
Without replacement	Selecting distinct units within a sample

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from itertools import combinations

def sample_means(population, size):
    if type(size) is not int or not 1 <= size <= len(population):
        raise ValueError('sample size')
    return [sum((population[i] for i in indices)) / size for indices in
combinations(range(len(population)), size)]

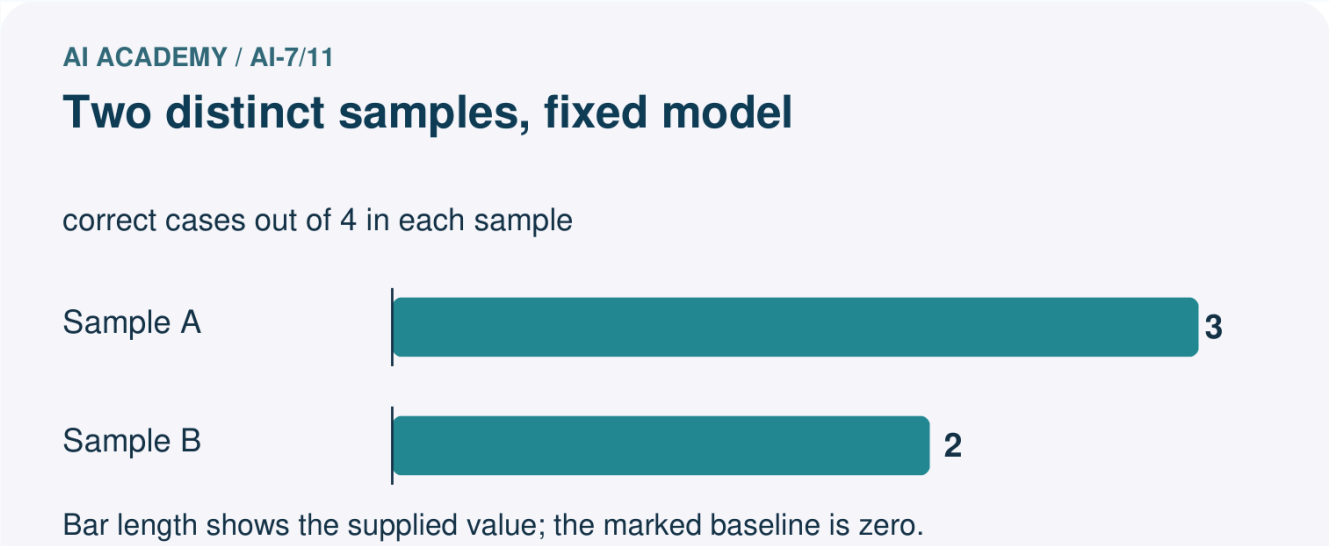
print(sample_means([1, 1, 3, 3], 2))
```

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sampling variability

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/11

Sampling variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep model fixed

No training or threshold change occurs between samples.



2. Compute observed rates

A is 75%; B is 50%, differing by 25 percentage points.



3. Consider sampling

Different included cases can change a small-sample rate.

CHANGE ONE THING

Evaluate the exact same fixed four cases twice with deterministic predictions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Sampling variability**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: This finite example is not a general normality guarantee or a model of a biased convenience sample.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate the two four-case rates and their difference. Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

No **training** or threshold change occurs between samples. A is 75%; B is 50%, differing by 25 percentage points. Different included cases can change a small-sample rate.

Changed case: The counts should agree unless some condition, data or scoring changed. The sampling explanation no longer applies when the records are identical.

Changed-case reasoning: The counts should agree unless some condition, data or scoring changed. The sampling explanation no longer applies when the records are identical.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main six uniform ID pairs yield means 1, 2, 2, 2, 2, 3, with probability 4/6 for mean 2. Picture 75%/50% rates cannot borrow that probability or a confidence interval; identical deterministic records should replay consistently. Selected values 0, 0, 4, 4 with distinct IDs yield means 0, 2, 2, 2, 2, 4, averaging 2; mean 2 occurs four of six times.

Preserve the uniform without-replacement ID sampling rule rather than counting unique-value pairs.

Diagnose the current misconceptions

The worked result establishes every future case: This finite example is not a general normality guarantee or a model of a biased convenience sample.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Population values are 0, 0, 4, 4, with four distinct IDs. Choose two IDs uniformly without replacement. Enumerate all six unordered pairs, calculate their sample means and find their average.

Teacher explanation: Means 0, 2, 2, 2, 2, 4; average 2; mean 2 occurs 4/6. Equal values have distinct IDs and count as separate eligible units.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Sampling variability** means different eligible samples can give different estimates even under the same population. Separate this from procedure changes. Consider finite population values 1, 1, 3, 3 with distinct IDs. Enumerate the six unordered samples of size 2 **without replacement**. Their means are 1, 2, 2, 2, 2, 3.

W2 Trace the worked case

Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2. Show the intermediate steps and give the

conclusion with its scope.

Teacher explanation: Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ.

W3 Audit the limit

A report omits this qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: This finite example is not a general normality guarantee or a model of a biased convenience sample.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. This finite example is not a general normality guarantee or a model of a biased convenience sample.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ. Qualification: This finite example is not a general normality guarantee or a model of a biased convenience sample. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Population values are 0, 0, 4, 4, with four distinct IDs. Choose two IDs uniformly without replacement. Enumerate all six unordered pairs, calculate their sample means and find their average.

Teacher explanation: Means 0, 2, 2, 2, 2, 4; average 2; mean 2 occurs 4/6. Equal values have distinct IDs and count as separate eligible units.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Means 0, 2, 2, 2, 2, 4; average 2; mean 2 occurs 4/6. Equal values have distinct IDs and count as separate eligible units. Require intermediate reasoning and the stated scope. Fresh retry: Population2, 2, 6, 6 under same sampling. Expected reasoning: Means2, 4, 4, 4, 4, 6; average4; mean4 probability2/3. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **sampling variability evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **sampling variability**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Uncertainty intervals. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/12, ai-7/14, ai-7/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **sampling variability**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

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>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Population values are 1, 1, 3, 3, with four distinct IDs. Choose two IDs uniformly without replacement. List all six unordered pairs and their means, then find the mean of those means and the chance of obtaining a mean of 2. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Mean = (1+2+2+2+2+3)/6=2; mean 2 occurs 4/6=2/3. The population mean is 2, while individual samples differ.

Check every learner before discussion: Use the worked case for **Sampling variability**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: This finite example is not a general normality guarantee or a model of a biased convenience sample.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Evaluate the exact same fixed four cases twice with deterministic **prediction**s. Require a **prediction** and a reason before comparing.

Expected result: The counts should agree unless some condition, data or scoring changed. Reason: The sampling explanation no longer applies when the records are identical.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

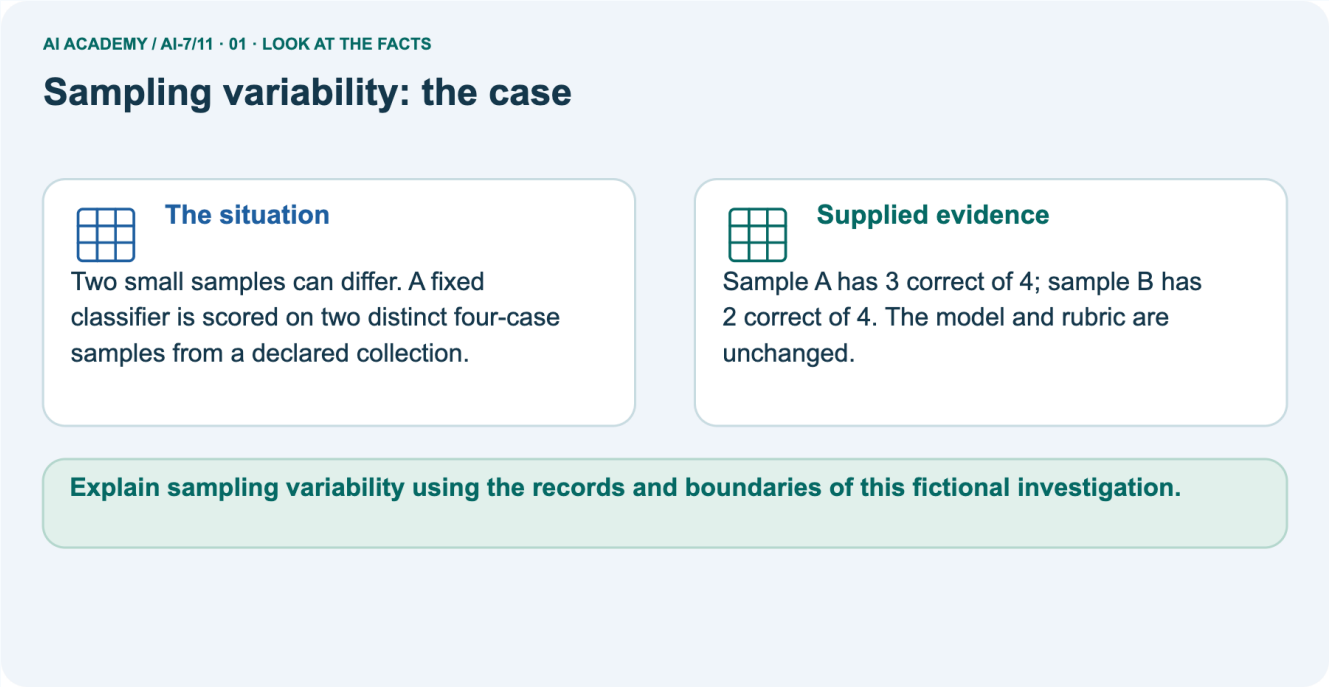
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Sampling variability: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

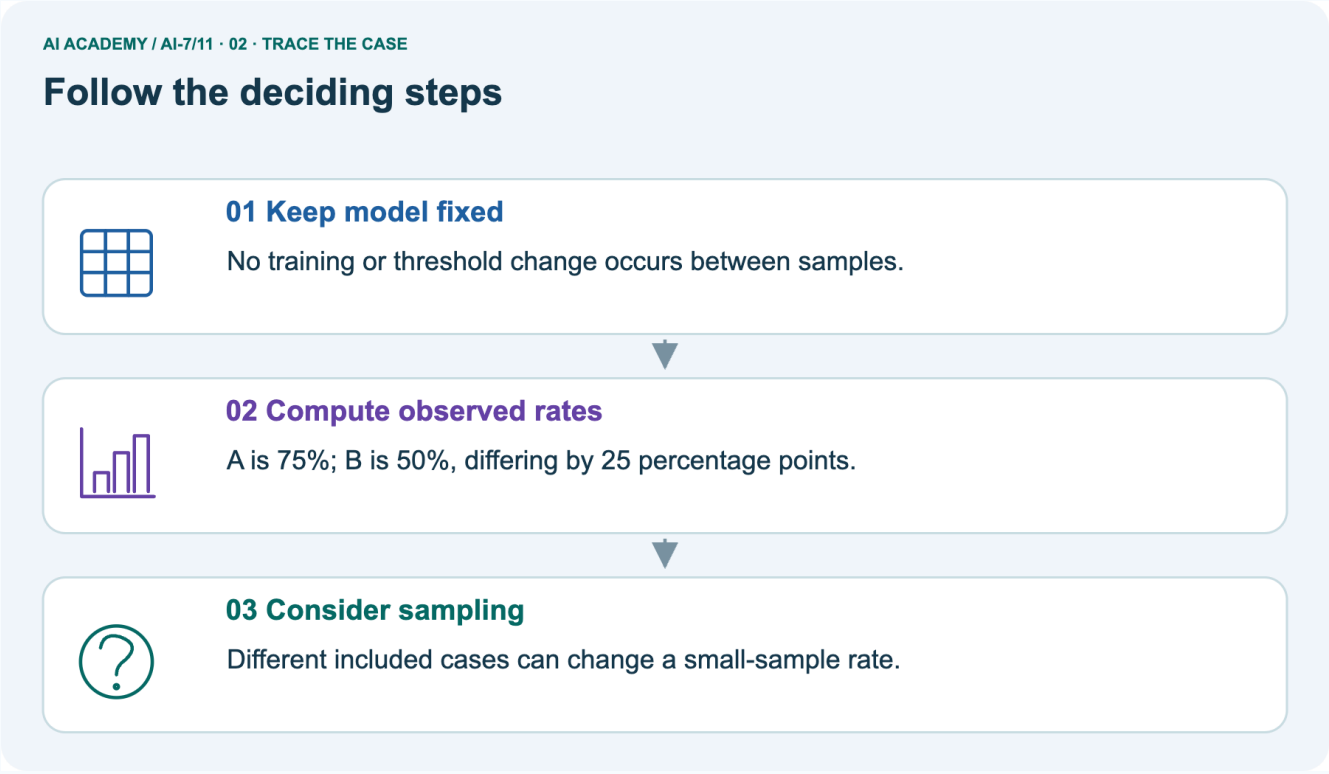
Two small samples can differ. A fixed classifier is scored on two distinct four-case samples from a declared collection. Supplied facts: Sample A has 3 correct of 4; sample B has 2 correct of 4.

The model and rubric are unchanged.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



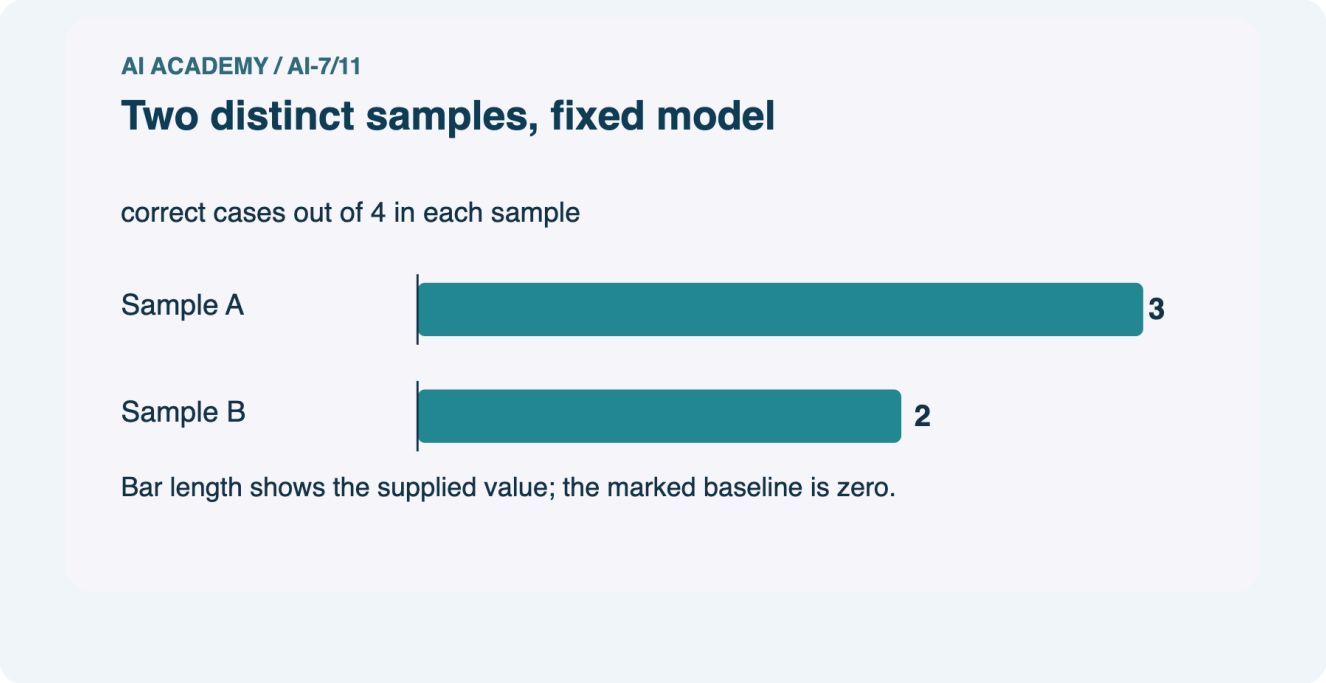
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep model fixed: No **training** or threshold change occurs between samples. → Compute observed rates: A is 75%; B is 50%, differing by 25 percentage points. → Consider sampling: Different included cases can change a small-sample rate.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two distinct samples, fixed model



Follow the labelled picture with the matching explanation. Open the vector diagram

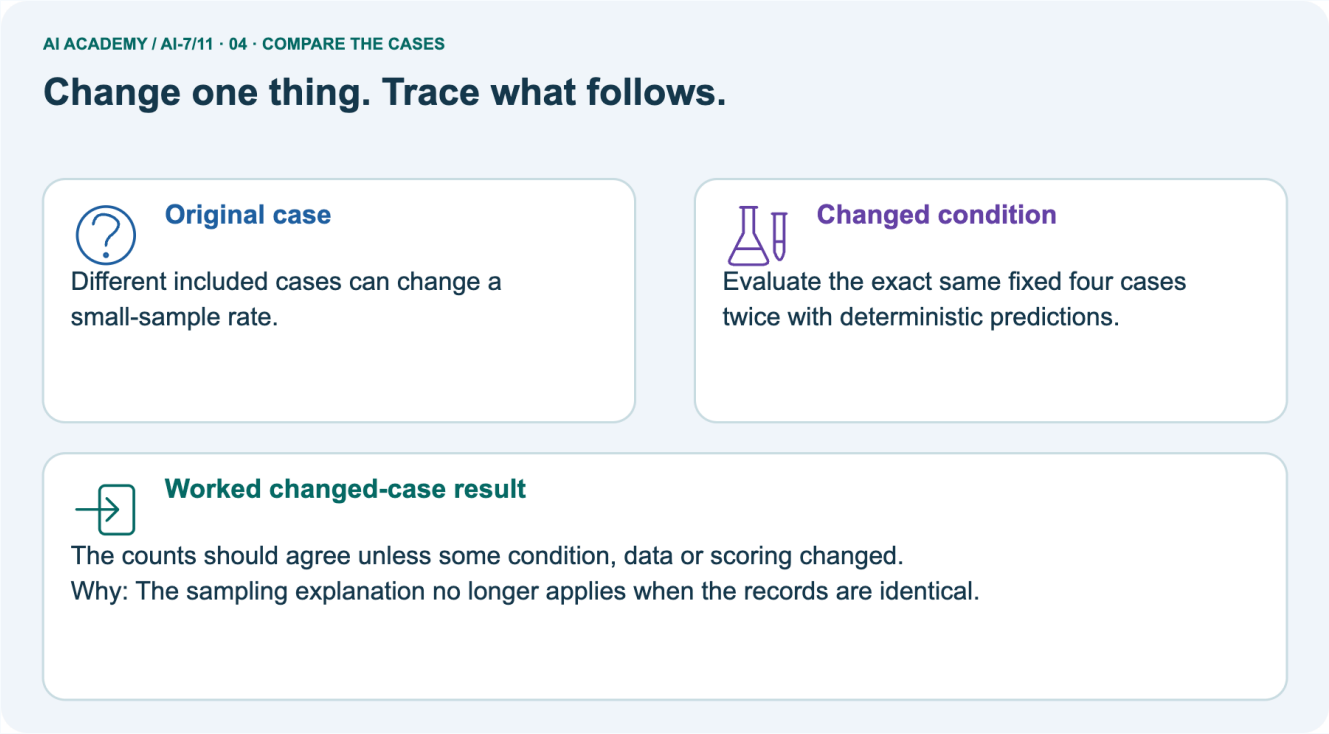
Two distinct samples, fixed model. Units: correct cases out of 4 in each sample. Sample A: 3. Sample B: 2.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Different included cases can change a small-sample rate. Changed condition: Evaluate the exact same fixed four cases twice with deterministic **prediction**s.

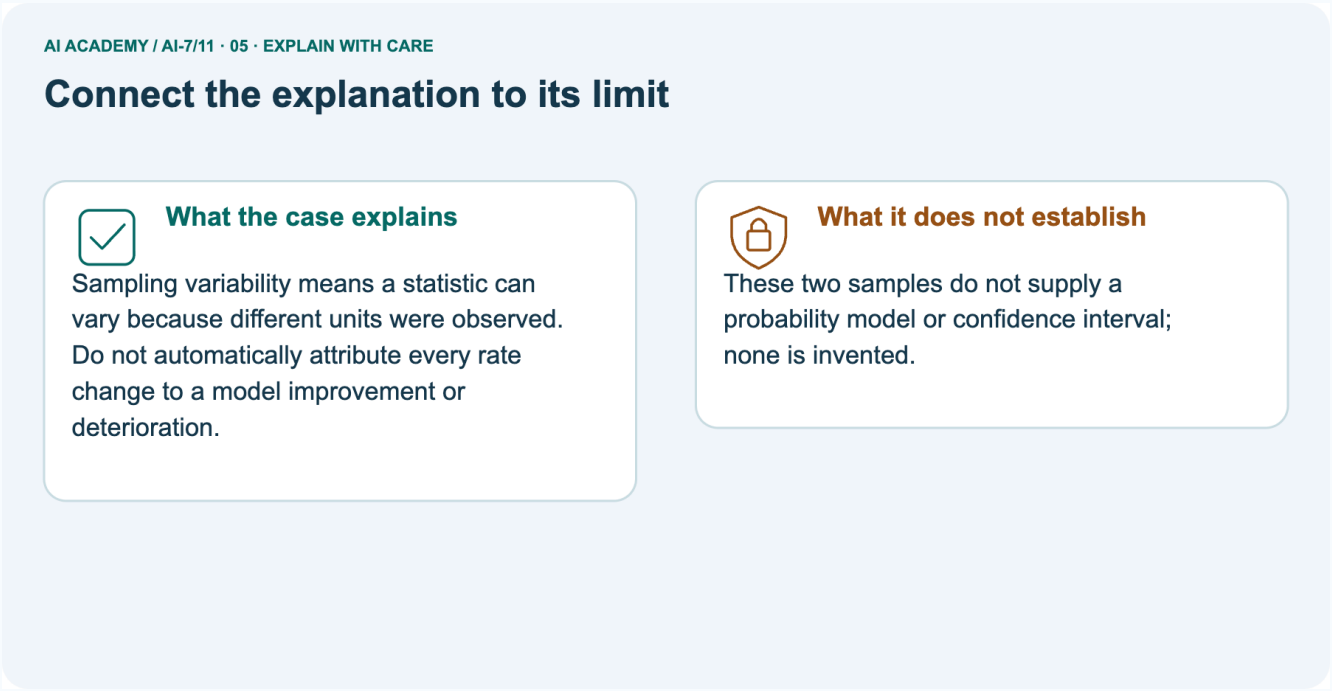
Expected result: The counts should agree unless some condition, data or scoring changed. Reason: The sampling explanation no longer applies when the records are identical.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The sampling explanation no longer applies when the records are identical.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Sampling variability means a statistic can vary because different units were observed. Do not automatically attribute every rate change to a model improvement or deterioration.

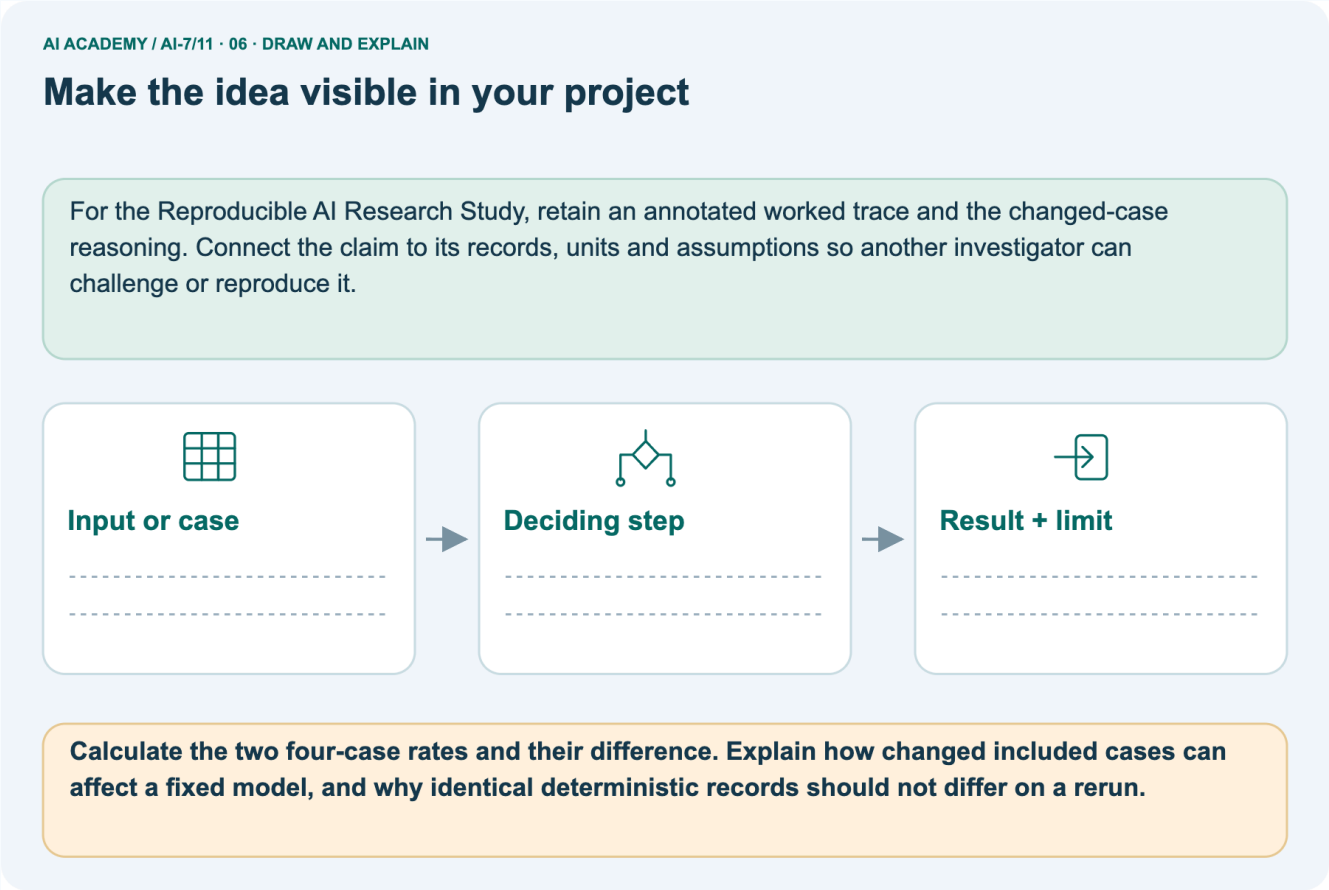
Limit: These two samples do not supply a probability model or confidence interval; none is invented.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These two samples do not supply a probability model or confidence interval; none is invented.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the two four-case rates and their difference.

Explain how changed included cases can affect a fixed model, and why identical deterministic records should not differ on a rerun.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No **training** or threshold change occurs between samples. A is 75%; B is 50%, differing by 25 percentage points. Different included cases can change a small-sample rate.

Changed case: The counts should agree unless some condition, data or scoring changed. The sampling explanation no longer applies when the records are identical.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sampling variability: the case

02. Follow the deciding steps

03. Two distinct samples, fixed model

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.